NTI

Academic Success & Intervention

# Student Academic Health Analysis

A full end-to-end data analysis pipeline examining 12,156 students across five grade levels over a complete academic year.


| | |
|---|---|
| Prepared by | **Nour Hatem** |
| Program | **NTI — Data Analysis Track** |
| Audience | **ENG. DONIA — Evaluation Committee** |
| Period | **March 2024 – March 2025** |


SQL Server 2025 → Python 3.12 → Power BI

# Pipeline Architecture

How raw data became actionable insight

## 01 Raw Data

5 CSV files

students · attendance · homework · performance · communication

→

## 02 SQL Server

Storytelling Phase

6-chapter narrative · Quality flags · Business questions

→

## 03 Python

Cleaning Phase

Normalization · Feature engineering · Export

→

## 04 Power BI

Dashboard Phase

6-page interactive dashboard

Total Records Processed

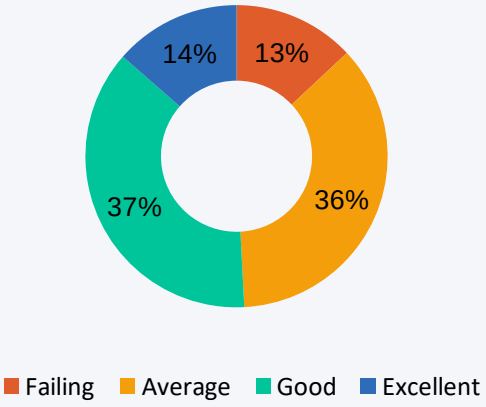
| 12,156 | 364,680 | 60,780 | 36,468 | 24,312 |
|---|---|---|---|---|
| Students | Attendance | Homework | Performance | Communication |

# Executive Summary

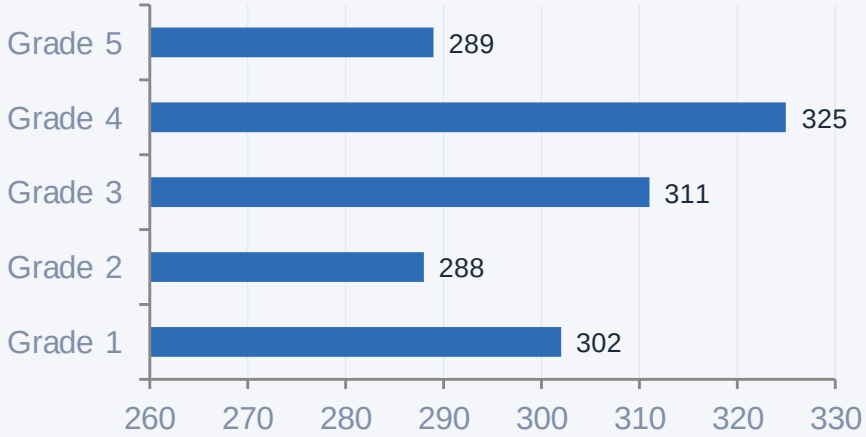
School health at a glance

| 12.156K | 74.96 | ~2K | 25.12% | 49.99% |
|---|---|---|---|---|
| Total Enrollment | Avg Academic Score | At-Risk Students | Attendance Rate | HW Completion Rate |

## Score Band Distribution

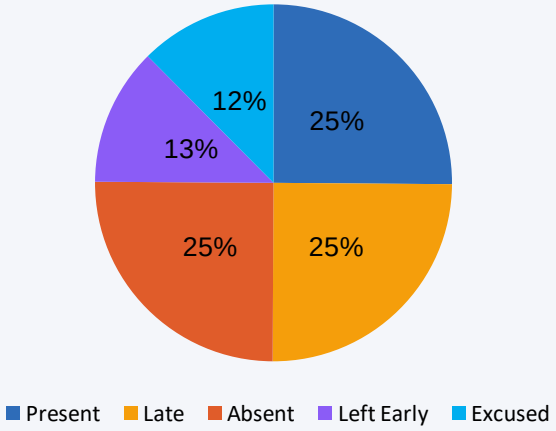


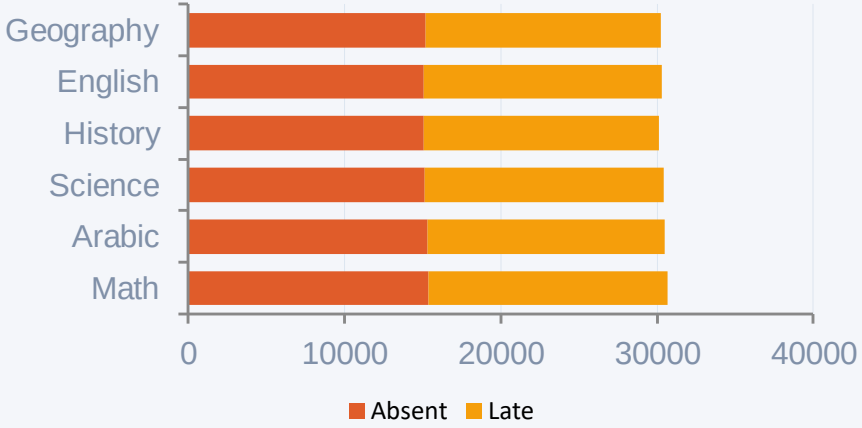

## At-Risk Count by Grade



# Attendance Analysis

Who is showing up, and when?



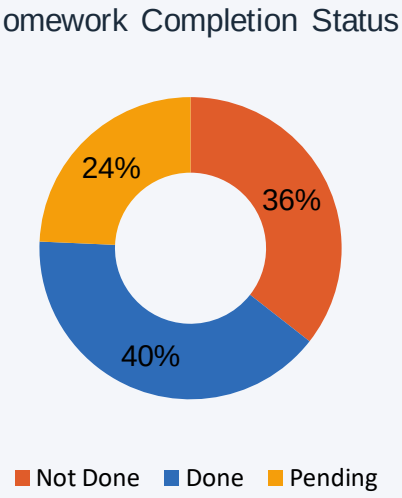



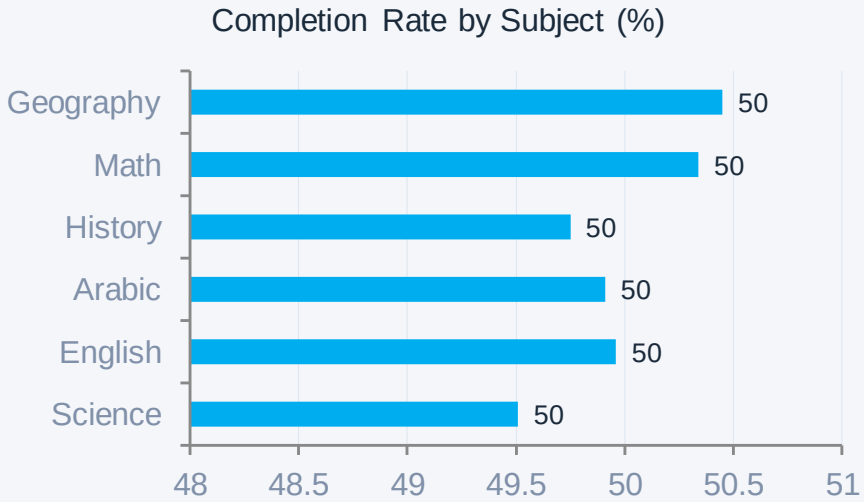

**Key Finding**

**Only 25.12%** of all records are marked Present — students are fully present in just 1 in 4 sessions. Late arrivals (24.96%) are as frequent as absences (~25%). **Absence is general, not subject-specific.**

# Homework & Engagement

Are students doing the work?



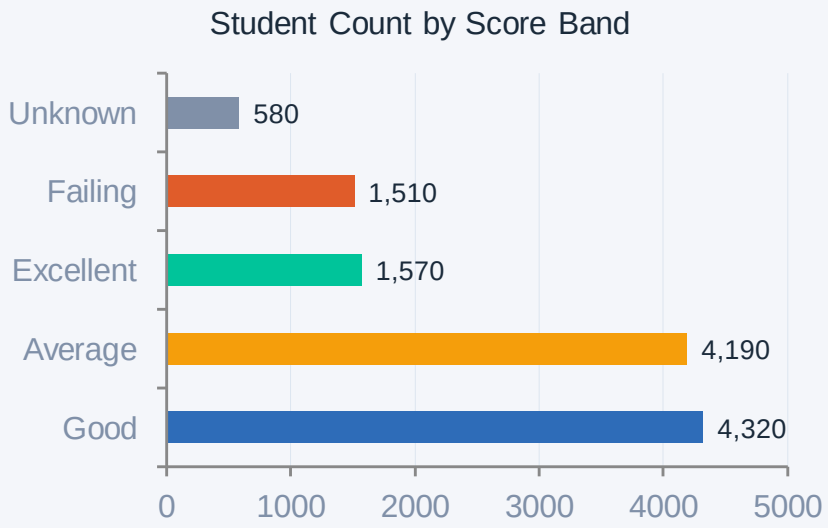

**Key Finding**

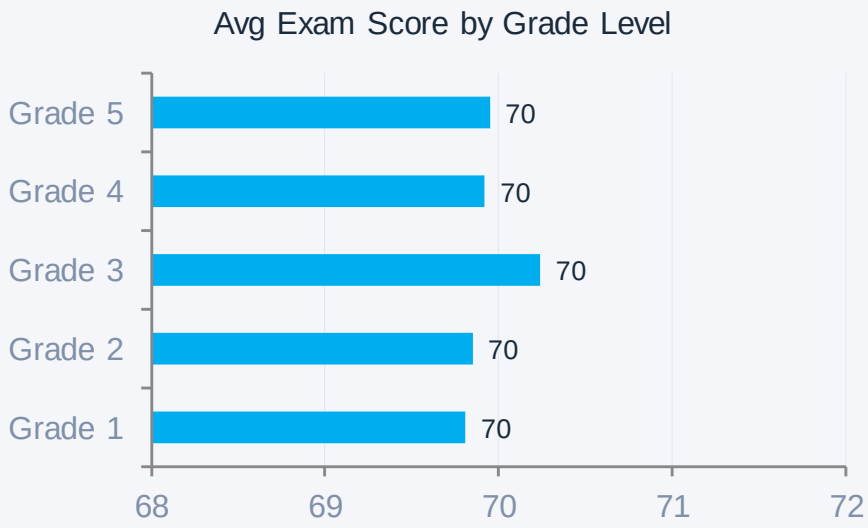
**1 in 3 assignments is never completed.** Science has the lowest completion rate. Guardian-signed assignments show marginally higher completion — suggesting parental involvement has some effect but is not the primary driver of engagement.

# Academic Performance

How are students scoring?

| 74.96 | 40 – 110 | 12.87% | 12.41% |
|---|---|---|---|
| School Avg Score | Score Range | High Achievers | Failing Rate |



# Parent Communication

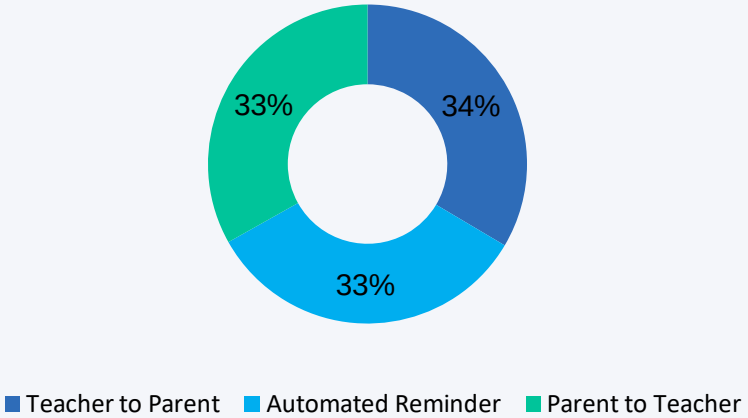
Is the school connected to parents?

**24K**
Total Messages

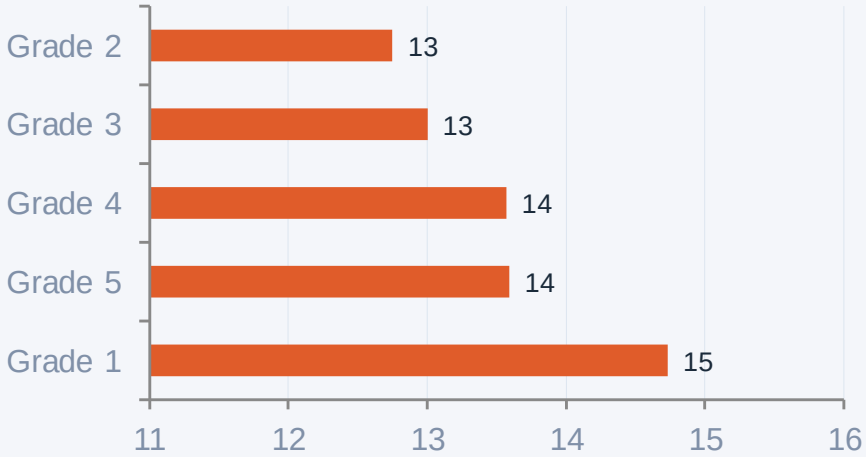
**1,644**
Silent Students

**13.5%**
Communication Gap

## Message Type Distribution



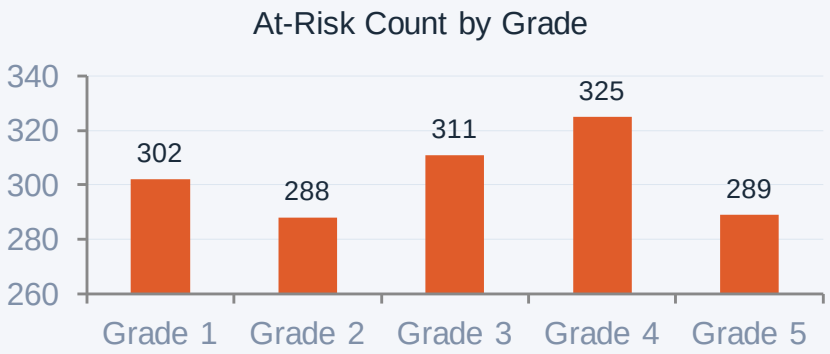

## Silent Students % by Grade



# At-Risk Intervention

Who needs action before the year ends?

| ~2K | 12.41% | 54.06 | 25.27% |
|---|---|---|---|
| At-Risk Students | % of School | At-Risk Avg Score | At-Risk Attendance Rate |




**At-Risk Flag Criteria**

| | |
|---|---|
| Attendance Rate | < 20% |
| Avg Exam Score | < 60 |
| Missed Assignments | > 3 |

All 3 conditions must be true simultaneously

**Most at-risk students have parent communication records.** Communication is happening but not translating into improvement. The intervention strategy needs to change — not just its frequency.

# Data Quality Issues & Fixes

What we found and how we resolved it

| | Issue | Fix |
|---|---|---|
| **Date_of_Birth**<br>Two date formats (ISO & US) | → | **pd.to_datetime(format='mixed')** |
| **Attendance_Status**<br>Casing inconsistencies + 'absnt' typo | → | **str.upper() + map + ABSNT→ABSENT** |
| **homework.Status**<br>Emoji + text variants mixed | → | **Set-based mapping → Done \| Not Done \| Pending** |
| **Guardian_Signature**<br>Imported as BIT — blanks not NULL | → | **CAST in SQL + fillna(False) in Python** |
| **Homework_Completion_%**<br>-5 invalid values + mixed % suffix | → | **Strip %, cast float, -5 → NaN, subject median fill** |

# Recommendations

What the data is telling us to do

## 01 Deploy the Intervention List

Use the At-Risk page as a live working tool. Each row is a student requiring a direct conversation before year-end.

## 02 Fix Grade 1 Communication Gap

14.73% of Grade 1 students have zero contact. Youngest students need the most parental engagement — build a structured onboarding protocol.

## 03 Rethink Communication Strategy

~200 at-risk students have active contact but declining performance. Volume is not the problem — quality and follow-through are.

## 04 Fix Data at Entry Point

Attendance typos, emoji status fields, and mixed date formats indicate no input validation. Standardize at source before the next year.

## 05 Build a Trend Detection Model

Add individual-level attendance trend monitoring. Flag students before they cross the at-risk threshold, not after.

## 06 Subject-Level Performance Model

Determine whether a student's low score is isolated to one subject or signals a broader academic decline across all subjects.

# Thank You

## NTI Academic Success & Intervention

**Portfolio**

nourhatem.web.app

**LinkedIn**

linkedin.com/in/nour-hatem-

**Kaggle**

kaggle.com/nourelhashash

**GitHub**

github.com/nour-hatem

| 12,156 | ~2K | 6 | 3 |
|---|---|---|---|
| Students | At-Risk | Dashboard Pages | Pipeline Phases |